Forecasting Average Month-Ahead European Gas Prices and Price Volatilities, 2015–2026

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Abstract

This paper assumes perfect foresight, one month ahead, to test the forecastability of the European natural gas price (the Dutch TTF benchmark), using a monthly dataset spanning January 2015 to May 2026. First, it tests whether and how much TTF can be forecast from European gas market fundamental variables: gas market balance, storage, power generation, pipeline supply, and weather. Second, global gas variables are added - global LNG supply and trade, non-European weather - and the paper asks whether the addition improves on the European model to forecast TTF prices. Third, it models TTF price volatility. Fourth, machine learning techniques are used as robustness checks on level results.

The paper benchmarks results against a random walk, assumes perfect foresight of independent variables and checks results with regularisation and dimensionality reduction. The average month-ahead TTF price is close to a random walk. Only storage change and price momentum carry signals, and global fundamentals contribute little. Machine learning checks support this. Structure does however exist. TTF volatility is modestly forecastable out of sample, even without perfect foresight. While this paper's volatility result merely confirms the literature, it also contributes a new result to the literature: next month's average European gas price is difficult to forecast even with perfect foresight.

Keywords: natural gas, TTF, cointegration, GARCH

1. Introduction

European natural gas prices have become much more important, particularly since the start of the Ukraine war. The Title Transfer Facility (TTF) price massively rose and collapsed again as Europe lost the bulk of its Russian pipeline supply and instead sourced gas through the global LNG market. That episode makes the European gas prices difficult to study: a single crisis dominates any recent sample.

This paper asks whether the month-ahead change in TTF can be forecast using a January 2015 to mid-2016 dataset. The horizon is fixed at one month (H = 1) throughout, to maximise degrees-of-freedom. Every forecast is evaluated by a walk-forward expanding-window against a random walk with a Diebold–Mariano test. The study assumes perfect-foresight or contemporaneous information (aside from the price itself) to forecast TTF prices.

Section 2 provides a literature review. Section 3 provides a description of data and methods. Sections 4–7 provide details of the four parts of the study. Section 4 builds a European gas model. Section 5 widens the variable set to include global natural gas variables. Section 6 examines TTF price volatility. Section 7 tests machine-learning methods as a robustness check on level results. Sections 8 and 9 discuss results and conclude.

2. Literature review

2.1 Price forecasting. Storage and weather often feature in studies covering the short-term dynamics of natural gas prices. Structural-VAR studies for NCG (Nick & Thoenes, 2014 using weekly data) and Henry Hub prices (Wiggins & Etienne, 2017 using weekly data) support this. Martínez & Torró (2023) find a weak but significant influence of storage on NBP essentially using OLS on monthy data. That weak-storage finding is a close match to this study's muted result. On predictability, Baumeister et al. (2024) show Henry Hub prices are forecastable up to two years in the future using real-time inputs, including futures prices, in a pool of forecasting models. This paper uses a pool of models, including autoregressive models, on monthly data. The finding for the H = 1 mean shows that TTF prices are difficult to forecast. This is a new, though negative, contribution to the literature.

2.2 Volatility and the crisis. Research focussing on TTF volatility is limited but has grown since the European energy crisis. Berrisch & Ziel (2022) obtains month-ahead European gas forecasting gains that are essentially distributional (volatility and tails), not price forecasting improvements, using state-space and volatility modelling. Botta, Cerqueti and Savona (2025) use daily TTF data but focus on conditional volatility and its transmission, estimating a suite of ARMA-GARCH specification and produce out-of-sample TTF volatility forecasts. This paper's results confirms that there is structure to TTF volatility.

2.3 Machine learning techniques. A growing machine-learning literature forecasts European gas prices at high frequency. Bajatović, Erdemlioglu & Gradojević (2024) forecast day-ahead TTF and Henry Hub prices with non-linear, nonparametric deep-learning models on daily data, reporting gains over parametric benchmarks. Böhm et al. (2023) apply feed-forward artificial neural networks to forecast day-ahead German (NCG) gas and EU carbon prices on roughly thirteen years of daily observations (2007–2020, ~4,500 data points). Both use daily multi-year samples of several thousand observations in contrast to the monthly, small number of observations in this study. This paper nevertheless applies robustness checks using ML to test month-ahead price level forecasting results. The paper finds these ML techniques do not find significant results for either the European or global price level forecasting exercises.

3. Data and methodology

3.1 Data. This study's dataset is monthly January 2015 to May 2026 (~137 observations). Front-month TTF prices are in USD/mmbtu, obtained from Yahoo Finance. European gas market fundamentals cover EU+UK gas balance and storage; production, net pipeline imports, LNG imports, net supply, total gas demand; power demand and power generation (gas, coal, nuclear, hydro, residual load); Norwegian pipeline supply (production, supply reduction, planned and unplanned outages); and European weather (heating and cooling degree days, wind speed, solar irradiation, Nordic precipitation). European data was collected from AGSI, Eurostat, national statistical agencies, ENTSOG, JODI, ENTSOE, Ember, Gassco, ECMWF, and miscellaneous sources. The global gas model adds non-European weather and supply risk (US and Northeast Asia degree days, Atlantic hurricane energy, Gulf storm days); global LNG supply and trade (world nameplate capacity and capacity offline); Asia and India LNG imports; Qatar, Australia, US, Southeast Asia, and Nigeria LNG exports; and financial variables (VIX and USD–EUR FX). Global data was compiled from ECMWF, JODI, IGU, GIIGNL, IEA, and miscellaneous sources.

All price series are in nominal terms. Because the horizon is one month and the target is a log return, deflation would subtract only the monthly inflation rate, which is negligible. Adopting real prices would moreover require justifying a particular deflator (e.g. US CPI, a PPI, or a global aggregate), so prices are kept in nominal terms.

3.2 No-look-ahead feature construction. Every predictor was transformed into a deseasonalised anomaly series: the raw series minus an expanding, prior years-only calendar month average. Variables that fail stationarity tests in levels are entered as month-over-month changes, which are stationary. Predictors are then either lagged one month (in a lagged, real-time forecast) or entered contemporaneously. Own-return momentum is always lagged.

3.3 Evaluation discipline. Model results are estimated with Newey–West (HAC) standard errors and screened by ADF for stationarity.

Log TTF prices test as non-stationary (I(1)) over the full sample. Augmented Dickey-Fuller (ADF) tests result in p = 0.46 for TTF. It has a unit root in levels and is stationary in first differences.

Specifications are selected using only pre-January-2025 data, i.e. the training data set. Forecasts are evaluated by a walk-forward expanding window that refits each month, benchmarked against a random walk and the historical mean, with significance judged using a Diebold-Mariano test with Newey-West variances.

Coefficients and transmission structure are re-estimated in a crisis regime versus calm or normal conditions. The crisis window is imposed on economic grounds at the onset of the energy crisis (September 2021 to July 2023). The high- versus low-volatility split addresses potential lookahead bias by using the trailing six-month return volatility relative to its expanding median. Because an imposed break date is a choice, the study corroborates the split with a Gregory-Hansen (1996) test that estimates a single break endogenously, and the data confirm the choice: both the level-shift (Model C) and regime-shift (Model C/S) specifications place the break at September 2021, and both reject the no-cointegration null at the 1% level.

3.4 The perfect foresight mode. The study runs each mean model in a mode in which same-month fundamentals enter contemporaneously. It assumes the modeller forecasts this month's storage, weather, and trade when predicting this month's price one month ago. This is an explanatory upper bound.

4. European gas modelling

4.1 The 'Core' model, i.e. the lagged model. With every predictor lagged, only four of ~27 which were found stationary under ADF+KPSS tests were individually significant at 5% using HAC estimators in the training window. Of these four predictors, none clear a Bonferroni family-wise error rate (FWER) significance, and storage change (coefficient −0.017, t = −2.67, p = 0.008) and momentum (coefficient +0.29, t = 2.54, p = 0.011) together reach the lowest Benjamini-Hochberg false discovery rate (FDR) at ~15%. The FDR would probably be even higher under a Benjamini-Yekutieli FDR which accounts for cross-correlated predictors. LNG-imports change is individually significant ($p \approx 0.03$–0.04) but on a regime split can become significant, e.g. in a 2021-2022 crisis window as tested using a joint Wald (p = 0.002).

Despite the statistical limitations noted above, the study tested the following model (the 'Core' model), which appears to be the best candidate given the data:

$$\Delta \log \mathrm{TTF}_t \;=\; \beta_0 \;+\; \beta_1\, \tilde{r}_{t-1} \;+\; \beta_2\, \Delta\tilde{S}_{t-1} \;+\; \beta_3\, \tilde{H}_{t-1} \;+\; \varepsilon_t$$

where $\Delta \log \mathrm{TTF}_t$ is the one-month log return of front-month TTF (the H = 1 target); $\tilde{r}_{t-1}$ is lagged price momentum (the own log-return anomaly); $\Delta\tilde{S}_{t-1}$ is the lagged EU+UK *storage-change* anomaly; $\tilde{H}_{t-1}$ is the lagged European HDD anomaly; and ε_t is the error. A tilde denotes a deseasonalised anomaly (the raw series minus an expanding, prior-years-only calendar-month climatology), and every predictor is lagged one month so the forecast uses only information available at the forecast origin. Coefficients are estimated by OLS with Newey-West (HAC) standard errors and forecasts are formed by an expanding window walk-forward.

The above tested model's forecast errors neither significantly beats a random walk (full-window OOS Campbell-Thompson $R^2 \approx +5.7\%$, Diebold-Mariano 0.58) nor survives compared to a random walk on test data (negative OOS Campbell-Thompson R^2 post-2025). So, a European model at most may deliver storage plus momentum and nothing else that survives statistical scrutiny, and even that does not beat a random walk at H = 1 on the test data. The tested model is therefore not statistically distinguishable from noise.

4.2 Conditional models. Granting perfect foresight of every same-month fundamental variable approximately doubles in-sample fit to $R^2 = 0.42$ and beats the random walk over the full sample (Diebold-Mariano 2.36). This is the 'Conditional I' or 'unrestricted foresight' model. However, there are two problems with it. First, the strongest contributors, contemporaneous coal power generation (t = 4.4) and pipeline-imports change (t = 3.7), carry the wrong economic sign, likely reflecting reverse causality (endogeneity). Second, on the post-2025 test data the model's OOS Campbell-Thompson R^2 is −26%.

Dropping coal power generation and pipeline-imports change, the following 'Conditional II' model was estimated:

$$\Delta \log \mathrm{TTF}_t \;=\; \beta_0 \;+\; \beta_1\, \tilde{r}_{t-1} \;+\; \beta_2\, \Delta\tilde{S}_t \;+\; \beta_3\, \tilde{H}_t$$

$$+\,\beta_4\, \tilde{C}_t \;+\; \beta_5\, \tilde{W}_t \;+\; \beta_6\, \tilde{I}_t \;+\; \beta_7\, \tilde{P}_t \;+\; \varepsilon_t$$

where the target $\Delta \log \mathrm{TTF}_t$ is the one-month log return of front-month TTF; $\tilde{r}_{t-1}$ is lagged price momentum (the sole predetermined control); and the fundamentals enter contemporaneously (dated t) — this same-month timing is the "perfect foresight" that makes the specification a conditional rather than a lagged forecast. They are the storage-change anomaly $\Delta\tilde{S}_t$ (semi-endogenous, flagged) and the exogenous weather block: heating-degree-days $\tilde{H}_t$, cooling-degree-days $\tilde{C}_t$, wind speed $\tilde{W}_t$, solar irradiation $\tilde{I}_t$, and Nordic precipitation $\tilde{P}_t$. A tilde denotes a deseasonalised anomaly (raw series minus an expanding, prior-years-only calendar-month climatology); ε_t is the error, with coefficients estimated by OLS and Newey–West (HAC) standard errors.

The predictors in the conditional model above are those that price cannot cause (weather) plus probably does not (storage). It results in a conditional $R^2 \approx 0.214$, with every sign economically correct but only storage is individually significant (t = −2.35). Even with perfect foresight, this model does not beat a random walk (full-window OOS Campbell-Thompson R^2 −1.3%, Diebold-Mariano −0.14).

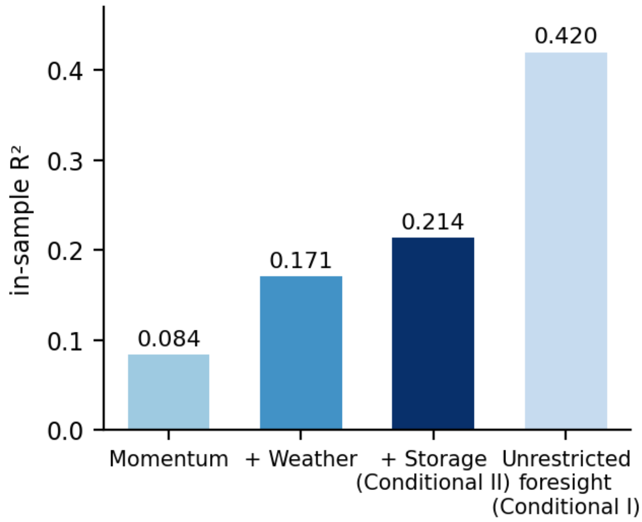

Figure 1. European in-sample explanatory upper bound. Storage plus exogenous weather under perfect foresight reaches about 0.21; the further rise to 0.42 under unrestricted foresight is largely simultaneity from adding endogeneous predictors.

4.3 The Lean Conditional model. The conditional model is over-parameterised with many predictors. To address this, a three-parameter conditional "lean" model (lagged momentum; contemporaneous storage and HDD) was estimated. Note this uses the same predictor set as the core model but allows perfect foresight of predictors (except the price). The results confirm the conditional model's failure is one of genuine signal weakness, not over-parameterisation. Both predictors are correctly signed and significant (storage t = −2.34; HDD t = +2.15). However, the model still does not beat a random walk (OOS Campbell-Thompson R^2 +4.0%, Diebold-Mariano 0.53) and remains marginally below the lagged core model (Diebold-Mariano −0.24).

4.4 Verdict on European gas modelling. Even granted perfect foresight, the models tested in this study do not beat a random walk.

Table 1 summarises the European out-of-sample results.

5. Global gas modelling

5.1 The global 'Compact' model. This model lacks perfect foresight of variables, as all variables are lagged. Widening the candidate pool to ~55 predictors (35 European, 19 global) and restricting predictors as mentioned below into the 'compact' model:

$$\Delta \log \mathrm{TTF}_t \;=\; \beta_0 \;+\; \sum_{j\in\mathcal{S}} \beta_j\, \tilde{x}_{j,t-1} \;+\; \varepsilon_t$$

where $\mathcal{S}$ is the compact predictor set the pipeline retained from the ~55 lagged candidates ($|\mathcal{S}| \leq 6$, momentum always included). The 6 is derived from the number of observations in the training set (106) and using the general rule that 10-20 observations per predictor is the minimum to avoid overfitting. Each $\tilde{x}_{j,t-1}$ is a lagged, deseasonalised anomaly; and $\mathcal{S}$ is chosen on the pre-2025 training window only, by an ADF+KPSS

Table 1: European out-of-sample results (H = 1, expanding walk-forward, versus a random walk). DM = Diebold–Mariano; RW = random walk.

European gas model	Full-window OOS R^2	DM vs RW	Test data OOS R^2
Core	+5.7%	0.58	−15.2%
Conditional I	+27.8%	2.36	−26%
Conditional II	−1.3%	−0.14	−24.1%
Lean conditional	+4.0%	0.53	−16.3%

stationarity gate, a univariate Newey–West t-ranking, collinearity pruning at $|R^2| > 0.70$, and retention of the top predictors significant at $p < 0.10$. Coefficients are estimated by OLS with Newey–West (HAC) standard errors.

The compact model's Diebold–Mariano statistic for the global model versus the European core model over the full data set window is essentially zero. On the post-2025 test data, the global model's OOS Campbell-Thompson R^2 versus a random walk is deeply negative ($\approx -59\%$). A model using a wider global data set adds noise, not signal.

5.2 The global conditional models. This adds global exogenous and global trade variables to the European lean conditional model. Granting perfect foresight of global variables lifts in-sample fit only modestly above the European conditional model. Adding the predictors piecemeal results in the following cumulative in-sample R^2:

The global variables do not significantly increase R^2 over the European lean conditional model. The global exogenous +0.04 increment concentrates in a single variable: NE-Asia degree days (coefficient +0.031, t = 2.21) is an Asian demand-pull channel, where cold Northeast Asian weather raises Asian LNG demand, diverts flexible cargoes away from Europe, and lifts TTF. The further rise to 0.298 is from adding the trade block, but those flows are probably partly endogenous as TTF prices influence trade.

Out of sample, the global conditional models do not provide statistically significant results. The global conditional model posts OOS Campbell-Thompson R^2 −12.2% (Diebold-Mariano −1.03 vs random walk; −1.08 vs the European conditional model), and adding the global trade block worsens it to −25.2%. Isolating the one significant global channel (NE-Asia degree days, added to the European lean) lowers OOS Campbell-Thompson R^2 from +4.0% in the European lean conditional model to +2.2% (Diebold-Mariano −0.58).

Table 3 summarises the global out-of-sample mean results.

6. Volatility

6.1 GARCH selection and persistence. Whereas it is difficult to forecast the TTF month-ahead price, there is more support for TTF month-ahead variance forecasting. Volatility clustering is confirmed before any model is fit.

$$\begin{aligned} r_t &= \log \mathrm{TTF}_t - \log \mathrm{TTF}_{t-1}, \\ r_t &= \mu_t + \varepsilon_t, \\ \sigma_t^2 &= \mathrm{Var}(\varepsilon_t \mid \mathcal{F}_{t-1}), \end{aligned}$$

$$\hat{\varepsilon}_t^2 = \gamma_0 + \sum_{i=1}^{q} \gamma_i \hat{\varepsilon}_{t-i}^2 + u_t,$$

Engle's ARCH-LM test of H_0: $\gamma_1 = \cdots = \gamma_q = 0$ (constant conditional variance) uses the statistic $T \cdot R^2 \sim \chi^2(q)$ and rejects decisively ($p = 0.0015$): the conditional variance is time-varying, which justifies testing GARCH-family models. GARCH(1,1) is

$$\sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2$$

Beyond GARCH(1,1), multiple specifications were tested. ARCH(1) is GARCH(1,1) with $\beta = 0$. The study then investigates asymmetric volatility using GJR-GARCH and EGARCH. The Student-t test adds a thick-tail parameter ν. Selecting on BIC $= -2 \log \hat{L} + k \ln T$, which charges $\ln T$ per parameter, GARCH(1,1) with normal errors wins.

The fitted process, $\sigma_t^2 = 10.58 + 0.277\, \varepsilon_{t-1}^2 + 0.723\, \sigma_{t-1}^2$, is IGARCH ($\alpha + \beta = 1.00$): shocks to volatility are effectively permanent, and the reaction coefficient $\alpha \approx 0.28$ is high — gas volatility both jumps sharply and persists. No significant asymmetry was found when GJR-GARCH and EGARCH were tested (gas lacks the negative-shock leverage found often with equities). The IGARCH is likely explained by regime changes.

6.2 Regimes. Conditional volatility averages ≈ 55% annualised (versus ~15–20% often for equities), ranging from a calm floor near 26% to a peak of 129% in early 2023, and is roughly twice as high in crisis as in calm (97% vs 47%).

6.3 Do fundamentals explain the variance? Regressing log conditional volatility on fundamental stress and a crisis dummy gives $R^2 = 0.45$. This is an order of magnitude more than fundamentals explained of the mean in calm markets. Let $\hat{\sigma}_t$ be the fitted GARCH(1,1) conditional volatility. Regressing its log on fundamental-stress magnitudes and a crisis dummy (OLS, Newey–West HAC standard errors),

$$\begin{aligned} \log \hat{\sigma}_t = \delta_0 + \underset{(t=1.81)}{\delta_1} |\Delta \tilde{S}_t| + \delta_2 |\tilde{W}_t| \\ + \underset{(t=8.7)}{\delta_3} D_t^{\text{crisis}} + u_t, \quad R^2 = 0.45, \end{aligned}$$

where $|\Delta \tilde{S}_t|$ is the storage-surprise magnitude, $|\tilde{W}_t|$ the weather-stress (absolute weather-anomaly) magnitude, and $D_t^{\text{crisis}} = \mathbf{1}\{\text{Sep 2021–Jul 2023}\}$. The crisis dummy carries most of the fit ($t = 8.7$), which is partly mechanical — the GARCH volatility is high across the crisis by construction, so a crisis dummy is partly explaining the series by itself. The result is that storage-surprise magnitude is marginally associated with higher volatility ($t = 1.81$, $p = 0.07$), while weather (δ_2) is statistically insignificant. Storage mattered slightly for the mean and slightly for the variance too.

Table 2: In-sample R^2 build-up for the global conditional models (each row adds its block to the row above).

Model	Predictors added	In-sample R^2	Gain
European momentum only	lagged own return	0.084	—
European lean conditional	storage + European weather	0.214	+0.130
Global exogenous	non-EU weather, global LNG offline	0.254	+0.040
Global trade	LNG import/export flows	0.298	+0.044

Table 3: Global out-of-sample mean results (versus a random walk). DM = Diebold–Mariano, RW = random walk.

Global model	Full-window OOS R^2	DM vs RW	vs European counterpart
Compact	≈ core	≈ 0	no gain (DM ≈ −0.01)
Conditional II + global	−12.2%	−1.03	worse (DM −1.08 vs Conditional II)
Lean conditional + global (+ NE-Asia DD)	+2.2%	0.27	worse than Lean conditional (DM −0.58)
ECM (core + error-correction)	+2.5%	0.27	worse (DM −1.07)

6.4 Out-of-sample variance forecasting. The GARCH(1,1) model beats naive benchmarks out of sample. On QLIKE, GARCH(1,1) beats a rolling-window forecast (Diebold-Mariano +1.62).

7. Machine learning checks on price level results

7.1 Rationale and constraints. With ~137 observations and up to ~55 predictors, the data is sparse and models based on this dataset could easily overfit. The machine learning (ML) techniques used here are completeness checks. The study tests the data using gradient-boosting, regularisation, and dimensionality reduction methods. All use z-score standardisation fit on training data only and hyperparameters are selected by time-series cross-validation, re-selected each walk-forward step.

7.2 Nonlinearity check. The time series econometric methods estimated above are linear. This part of the study tests machine learning for non-linearity and interactions among core predictors (lagged momentum, storage-change, HDD). The study tests a nonparametric route using a deliberately shallow, regularised gradient boosting (depth-2 trees, learning rate 0.05, subsampling). Gradient boosting merely reproduces the linear core (+0.12 vs +0.11 OOS R^2, Diebold-Mariano +0.20 — statistically zero) and fails the post-2025 slice (−0.27). The in-sample nonlinearity is therefore likely overfitting. Allowing nonlinearity adds little to forecasting.

7.3 Ridge regression. In the lagged (real-time) specifications, ridge regression does not beat the random walk (EU OOS Campbell-Thompson R^2 −0.049; global −0.064) and loses to the parsimonious core (DM −0.95 EU, −1.25 global) with large selected penalties (median $\lambda \approx 134$–203). Its largest standardised coefficients merely re-discover variables selected when estimating time series models above: storage, CDD, LNG imports, HDD. In the conditional (perfect-foresight) specifications, ridge regression posts a positive but insignificant OOS Campbell-Thompson R^2 (EU +0.104, DM 0.90).

7.4 Principal components regression. PCA was applied to the data, and while a dominant common factor was identified (the first principal component explains 16–22% of predictor variance, the first five 43–58%), it carries no month-ahead predictive power. The lagged PCR loses to the random walk and the core (EU PCR −0.033). Under perfect foresight, PCR posts the study's highest OOS numbers (global PCR +0.170, DM +1.47) but still statistically insignificant.

7.5 Machine-learning verdict. In lagged, real-time form, no ML model beats the storage-plus-momentum core or the random walk, and the signal is weak.

8. Discussion

The conditional mean $E(\Delta log(TTF_t|F_{t-1}))$ is close to a martingale, i.e. = $log(TTF_t)$. Its one durable, real-time predictor is storage change (and own-momentum). Global fundamentals, competing prices, weather, error-correction, latent factors, and even perfect foresight of the exogenous fundamentals do not reliably improve on "no change" out of sample. The little signal that exists is concentrated in crises, which are difficult to anticipate. Even under perfect foresight, no model would have beaten a random walk.

This study supports some volatility forecastability. The GARCH(1,1) model beats naive benchmarks out of sample. So, while TTF price forecasting models did not yield statistically signficant results, TTF price volatility forecasting did.

There are numerous limitations to this study. The number of observations is low (~137), which reduces statistical power and limits the number of independent variables. Further, the sample contains a single major crisis (the energy crisis, exacerbated by the start of the Ukraine war), which also reduces statistical power. The study is confined to H = 1 as longer horizons would lead to effectively reducing the available degrees of freedom and again limit statistical power.

9. Conclusion

Using a monthly 2015-2026 dataset under a uniform out-of-sample testing methodology, the study tested European and global natural gas market data, GARCH volatility, and ML for completeness. The result is consistent: the month-ahead TTF is difficult to forecast. European gas storage and price momentum are its only durable signals. Global information adds little even under perfect foresight. ML confirms the signals are weak. While this paper's volatility result merely confirms the literature, it also contributes a new result to the literature: next month's average European gas price is difficult to forecast even with perfect foresight.

10. References

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Appendix A. Reproducibility

The study is fully scripted and reproducible from a single monthly dataset (`NG_m_final.csv`). Data pipeline: `build_ng_dataset_monthly.py`. Main model: `ttf_monthly_paper_results.py`. All models fix H = 1, use no-look-ahead deseasonalised anomalies, and are evaluated by expanding walk-forward against a random walk with Diebold–Mariano tests.